CHAPTER 11

Integration

S59, S61, S63, S66

Key point · Objectives. Calculate and bound an integral; distinguish area from integral; use integration by parts; study a sequence of integrals.

Course notes

Area and antiderivative

For f continuous and positive on $[a,b]$, $a \leq b$, the integral $\int_a^b f(x)\,\mathrm{d}x$ is the area under the curve, above the axis, between $x = a$ and $x = b$, expressed in area units.

Key point · Fundamental theorem. If f is continuous on an interval I and $a \in I$, then $F_a(x) = \int_a^x f(t)\,\mathrm{d}t$ is the antiderivative of f that vanishes at a. For every antiderivative F,

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Every continuous function on an interval has antiderivatives. An integral is a number; an antiderivative is a function.

Proof in the positive increasing case. For $h > 0$, the rectangles give $hf(x) \leq F_a(x+h) - F_a(x) \leq hf(x+h)$. After dividing by h and taking the limit, the rate of change tends to $f(x)$. For $h < 0$, the increasing property gives $f(x+h) \leq f(t) \leq f(x)$ on $[x+h, x]$. Integrating,

$$(-h)f(x+h) \leq F_a(x) - F_a(x+h) \leq (-h)f(x).$$

Dividing by $-h > 0$, we obtain

$$f(x+h) \leq \frac{F_a(x+h) - F_a(x)}{h} \leq f(x).$$

Both bounds tend to $f(x)$. By continuity, $F_a' = f$. Every other antiderivative is $F_a + C$, hence the formula for evaluating between the bounds.

Properties

For continuous functions and bounds belonging to the same interval:

$$\int_a^a f = 0, \quad \int_b^a f = -\int_a^b f, \quad \int_a^c f = \int_a^b f + \int_b^c f, \quad \int_a^b (\lambda f + \mu g) = \lambda \int_a^b f + \mu \int_a^b g.$$

If $a \leq b$ and $f \leq g$, then $\int_a^b f \leq \int_a^b g$. In particular, if $m \leq f \leq M$, then $m(b-a) \leq \int_a^b f \leq M(b-a)$. For a function of arbitrary sign, the integral is defined through antiderivatives. The geometric area between the curve and the axis is $\int_a^b |f|$, obtained by splitting at the sign changes. Between two curves, it is $\int_a^b |f - g|$.

Example. $\int_{-1}^{1} x \, \mathrm{d}x = 0$, but the area between the line and the axis on this interval is 1.

Mean value and integration by parts

For $a < b$, the mean value is $\mu = \frac{1}{b-a} \int_a^b f(x) \, \mathrm{d}x$. The rectangle of height μ and width $b - a$ has the same algebraic area.

Key point · Integration by parts. If u, v are of class $\mathcal{C}^1$ on $[a, b]$,

$$\int_a^b u(x)v'(x) \, \mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, \mathrm{d}x.$$

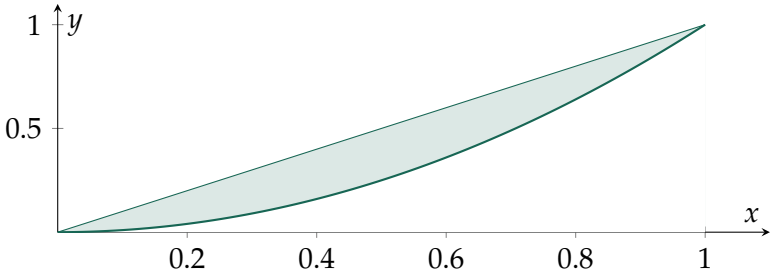
Proof. Integrate $(uv)' = u'v + uv'$, then isolate the required integral. The square bracket is indispensable.

Example. With $u = x$ and $v' = e^x$, $\int_0^1 xe^x \, \mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \, \mathrm{d}x = e - (e - 1) = 1$.

Rectangles and sequences of integrals

For f continuous and increasing on $[a, b]$, with $a < b$ and $n \geq 1$ and $h = \frac{b-a}{n}$, the sums $h \sum_{k=0}^{n-1} f(a + kh)$ and $h \sum_{k=1}^{n} f(a + kh)$ bound the integral. Their difference is $h(f(b) - f(a))$. For $I_n = \int_0^1 x^n f(x) \, \mathrm{d}x$, begin with the sign and comparisons of the integrand. If $0 \leq f \leq M$, then $0 \leq I_n \leq \frac{M}{n+1}$; the squeeze theorem gives $I_n \to 0$. An integration by parts can also provide a relation between I_n and I_{n+1}.

Convention at rank zero. In $I_n = \int_0^1 x^n f(x) \, \mathrm{d}x$, the expression x^0 denotes the constant function 1, including at the lower bound 0. Thus $I_0 = \int_0^1 f(x) \, \mathrm{d}x$; no indeterminate calculation is needed.



The area between $y = x$ and $y = x^2$ on $[0, 1]$ is $\frac{1}{6}$.

Research activity

Bounding an integral with rectangles — 35 min.

We want to bound $I = \displaystyle\int_0^1 x^2 \, \mathrm{d}x$. For an integer $n \geq 1$, we split $[0,1]$ into n intervals of equal length. Let L_n be the sum of the areas of the rectangles whose heights are the values of x^2 at the left endpoints, and U_n the sum obtained at the right endpoints.

1. For $n = 4$, draw the two families of rectangles, then calculate L_4 and U_4.
2. Justify that $L_4 \leq I \leq U_4$.
3. Write a Python function with argument n that returns L_n and U_n. Find $U_n - L_n$ algebraically.
4. Find the smallest integer n for which this bound has a width of at most 0.01.
5. Calculate the exact value of I using an antiderivative and check that it lies within the bound obtained for $n = 4$.

Progressive exercises

E1 · Practice. Calculate $\displaystyle\int_0^2 (3x^2 - 2x + 1)\ \mathrm{d}x$ by finding an antiderivative.

E2 · Practice. In an orthonormal frame, calculate the area of the region bounded by the curves $y = x$ and $y = x^2$, between the abscissae 0 and 1. Give the result in area units.

E3 · Practice. Calculate the mean value of the exponential function on the interval $[0, 2]$.

E4 · Practice. Using integration by parts, calculate $\displaystyle\int_1^e \ln x\ \mathrm{d}x$.

E5 · Practice. For every $n \in \mathbb{N}$, let $I_n = \displaystyle\int_0^1 \frac{x^n}{1+x}\ \mathrm{d}x$, with $x^0 = 1$ in the integrand. Prove that $0 \leq I_n \leq \frac{1}{n+1}$, then deduce the limit of (I_n).

Textbook or past-paper work

Barbazo, p. 260, exercises 28 and 29. In S66: exercise 28, stating explicitly the two functions chosen for the integration by parts. At home: exercise 29, to adapt this choice to a product containing a logarithm.

Homework

HW8 — A sequence of integrals — 60 min.

For every $n \in \mathbb{N}$, let $I_n = \displaystyle\int_0^1 x^n e^x \, \mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^x$.

1. Calculate I_0, then I_1 using integration by parts.
2. Prove that, for every $n \in \mathbb{N}$, $I_{n+1} = e - (n+1)I_n$.
3. By comparing the integrands, prove that (I_n) is positive and decreasing.
4. Show that $\frac{1}{n+1} \leq I_n \leq \frac{e}{n+1}$ for every $n \in \mathbb{N}$. Deduce the limit of (I_n).
5. In the calculation of I_{n+1} using the relation of question 2, by what factor is an error on I_n multiplied if the other rounding errors are neglected? Explain the risk of repeated calculation in this way.